

DEVA, Romania

WMO: 152300

Lat: 45.8650N

Lon: 22.8989E

Elev: 241

StdP: 98.46

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.1	-9.5	-15.7	1.0	-10.3	-12.7	1.3	-8.0	7.9	4.6	6.6	3.7	1.1	180	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	32.9	20.4	31.1	20.4	29.3	19.7	22.1	29.3	21.2	28.4	20.4	27.4	2.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.9	15.1	24.6	18.9	14.1	23.7	18.1	13.4	22.8	65.8	29.4	62.4	28.5	59.7	27.6	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.4	4.5	-16.2	35.2	3.9	2.3	-19.0	36.8	-21.3	38.2	-23.5	39.4	-26.4	41.1		
(5)			-16.4	23.8	3.9	1.3	-19.2	24.7	-21.5	25.5	-23.7	26.3	-26.5	27.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-0.8	1.7	6.1	11.8	16.0	19.6	21.3	21.4	16.6	11.2	5.4	0.6		
	DBStd	8.73	4.50	4.79	4.36	3.67	3.32	3.11	2.93	2.96	3.70	3.92	4.43	4.17		
	HDD10.0	1199	335	233	132	23	1	0	0	0	1	34	148	293		
	HDD18.3	2995	593	466	380	197	86	22	8	8	75	223	387	551		
	CDD10.0	1548	0	1	11	77	188	288	350	352	199	71	11	0		
	CDD18.3	302	0	0	0	1	14	61	100	102	24	1	0	0		
	CDH23.3	3128	0	0	0	33	185	601	1014	1012	267	16	0	0		
	CDH26.7	1114	0	0	0	5	29	186	394	418	80	1	0	0		
(14) Wind	WSAvg	1.6	1.4	1.6	1.9	1.9	1.7	1.7	1.7	1.5	1.6	1.4	1.4	1.5		
(15) Precipitation	PrecAvg	623	31	28	36	53	71	82	79	63	58	45	38	40		
	PrecMax	810	56	86	84	109	148	150	148	216	119	105	82	115		
	PrecMin	387	8	4	8	5	30	18	11	10	3	2	0	2		
	PrecStd	118	14	18	20	25	28	32	36	44	30	26	19	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	15.6	21.1	26.9	29.2	32.8	35.1	35.8	32.3	25.3	18.9	12.0		
		MCWB	7.5	9.5	11.4	15.3	18.0	21.4	20.7	20.2	18.7	15.9	12.9	8.9		
	2%	DB	8.2	12.4	18.3	23.4	27.3	30.7	32.8	33.1	28.7	22.7	16.1	9.4		
		MCWB	6.0	7.8	10.4	13.7	17.3	20.8	20.6	20.1	18.0	15.3	11.7	7.5		
	5%	DB	6.2	10.2	15.7	21.2	25.5	28.9	30.8	31.0	26.4	20.2	14.0	7.5		
		MCWB	4.8	6.7	9.1	12.5	16.9	20.0	20.5	20.3	17.4	14.1	10.7	6.1		
	10%	DB	4.6	8.2	13.4	19.1	23.5	27.0	28.7	29.0	24.1	17.8	11.8	6.0		
		MCWB	3.4	5.7	8.0	11.6	15.8	19.0	19.8	19.6	16.6	13.2	9.4	5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.1	10.0	12.5	16.2	19.9	23.5	23.2	23.6	20.8	17.5	13.7	9.5		
		MCDB	10.0	14.5	19.2	24.9	26.7	30.2	29.9	30.5	29.8	23.0	17.0	11.1		
	2%	WB	6.3	8.4	10.9	14.7	18.5	21.7	21.9	21.9	19.1	15.9	12.3	7.7		
		MCDB	7.7	11.6	16.9	22.1	24.9	28.6	29.3	29.2	25.9	20.7	15.1	9.0		
	5%	WB	4.9	7.2	9.7	13.5	17.5	20.6	21.1	20.9	18.1	14.8	11.1	6.2		
		MCDB	6.2	9.6	14.6	19.3	23.6	27.3	28.4	27.9	23.9	19.2	13.8	7.4		
	10%	WB	3.5	5.9	8.6	12.4	16.5	19.7	20.4	20.1	17.1	13.6	9.6	5.0		
		MCDB	4.5	8.3	12.8	17.5	22.4	25.8	27.3	27.2	22.5	17.2	11.6	6.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	8.1	10.3	11.7	11.7	12.0	12.9	13.1	11.6	11.0	7.9	5.4		
		MCDBR	8.0	11.7	14.7	15.9	15.4	15.2	16.7	16.8	16.1	15.1	11.1	7.8		
		MCWBR	5.7	7.3	7.7	7.2	6.6	6.2	6.0	5.6	6.3	7.8	6.8	5.7		
		MCDBR	7.4	10.7	12.9	13.7	13.6	13.9	14.4	14.2	13.5	13.3	10.0	7.4		
	5% WB	MCWBR	5.5	6.9	7.1	6.4	6.4	6.1	5.8	5.5	5.8	7.1	6.2	5.6		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.355	0.401	0.462	0.448	0.439	0.458	0.469	0.426	0.383	0.352	0.325		
	taud		2.326	2.247	2.182	2.041	2.144	2.215	2.168	2.132	2.217	2.329	2.366	2.360		
	Ebn at Noon		761	805	817	796	824	832	809	782	782	765	723	719		
	Edh at Noon		78	102	126	159	148	139	144	143	121	94	76	69		
(44) All-Sky Solar Radiation	RadAvg		1.28	2.07	3.19	4.33	5.16	5.78	5.68	5.20	3.70	2.58	1.51	1.04		
	RadStd		0.17	0.26	0.45	0.47	0.48	0.54	0.49	0.51	0.40	0.30	0.19	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.79	N/A	N/A	+1.46	N/A	N/A	N/A	-206	+167	+81
(47) Regional (70 neighbors)		+0.69	N/A	N/A	+0.93	N/A	N/A	-119	-182	+123	+62

Nomenclature: See separate page